

Brady Bangasser

Software Engineer

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Software engineer across the stack, from low-latency Go and Rust services and secure backends to the infrastructure and CI that ship them.

EXPERIENCE

System Administrator · Iowa State University

Apr 2025 – Present

Ames, IA

- Developed and tested a new feature in a customized SLURM scheduler in C, with Makefile-driven builds, debugging, and validation for stability and compatibility.

Head of Networking · Cardinal Space Mining

Aug 2024 – Present

Ames, IA

- Collected multiple camera streams and relayed them to a third-party platform for live judging.

Cloud Security Engineer · John Deere

May 2025 – Aug 2025

Johnston, IA

- Led a migration of IAM, security, and dev-management tooling from Python to Go, cutting runtime cost by over 10%.
- Deployed high-throughput Go services on AWS EKS to run compliance checks across the cloud estate.

Research Assistant — Machine Learning · Iowa State University

Oct 2024 – May 2025

Ames, IA

- Designed and trained object-classification models with PyTorch and OpenCV, tuning data preprocessing and tensor operations to cut training latency without losing accuracy.
- Probed model robustness by building adversarial exploit methods, then designed defenses to harden the models against them.
- Benchmarked accuracy and throughput across GPU configurations to guide training choices.
- Tested perception models on self-driving vehicles to catch false and failed classifications.

Lead Telemetry Engineer · Cyclone Rocketry

Aug 2024 – May 2025

Ames, IA

- Built modular software to control and acquire data from airframe instruments, improving reliability and maintainability.
- Wrote ground-station software to visualize live telemetry during flights.
- Designed a custom telemetry protocol for encrypted data routing over a mesh network, handling sensitive data across multiple interfaces at once.

Manager · Caribou Coffee

Oct 2021 – Jan 2025

White Bear Lake, MN

- Managed staff and daily operations, trained employees, and handled company devices and technical issues across the district.

PROJECTS

Omni HTTP Router

2025–2026

- Built a Rust HTTP router that resolves middleware at compile time into fully static stacks, removing traditional routing overhead.
- Benchmarked the static router against conventional middleware stacks to quantify the latency and allocation savings.

Tech: Rust, LLVM, Terraform

Computer Vision Parking Enforcement Radar

2025–2026

- Built a real-time computer-vision service that detects parking-enforcement vehicles and broadcasts locations to a user-facing app.
- Stored and queried detected-vehicle geolocation in PostgreSQL for fast spatial lookups.
- Architected a Redis Streams pipeline for low-latency, high-throughput image and detection routing from a REST API to processing servers, backed by PostgreSQL geospatial storage.

Tech: Python, Redis Streams, PostgreSQL, REST

Public Data Scraping and Processing

2025

- Wrote a compiler-like tool that pairs lexical analysis with an LLM to standardize and validate data formats, and trained an ML model to flag anomalies in the collected data.
- Rendered target sites with a custom engine and generated Lua/Python API wrappers to collect data reliably.

PortTorpedo

2025

- Built a decentralized, end-to-end encrypted online Battleship using UDP/TCP hole punching for peer-to-peer play, with cryptographic compression and integrity checks.
- Implemented UDP/TCP hole punching so peers connect directly with no central server.
- Added cryptographic integrity checks and compression over the peer-to-peer channel.

Tech: C++, C, AES, ECC, SHA3

Gin Gonic Routing & Middleware Framework

2025–2026

- Wrote a Rust tool that auto-generates and protects Go (Gin) routes by file path, with middleware for permission bits, request IDs, and structured logging.
- Generated per-route middleware for permission bits, request IDs, and structured logging.
- Cut boilerplate for standing up secure Go services with consistent, path-based configuration.

Tech: Rust, Go, Gin Gonic

Secure Backend Architecture & Authentication

2025–2026

- Built a secure backend with user authentication and JWT sessions over a fully encrypted MySQL database, exposing scalable, high-throughput API routes.
- Exposed scalable, high-throughput API routes with authentication enforced on every endpoint.
- Modeled and secured relational data in an encrypted MySQL schema with integrity constraints.

Tech: Go, MySQL, JWT

Lethal Company Sync

2023

- Built a multiplatform tool to sync files across devices using encryption, hashing, and compression.
- Synced only changed files using content hashing, and compressed transfers to cut bandwidth.
- Encrypted synced files end to end so data stayed protected across platforms.

Tech: C++, C, Gzip, AES, Bcrypt, RSA, SHA3

EDUCATION

M.S. Computer Science · Iowa State University

Aug 2025 – May 2027

GPA: 4.00

- Concurrent B.S./M.S. program

B.S. Computer Science · Iowa State University

Aug 2024 – Dec 2026

GPA: 3.79 · Dean's List · Minors in Cybersecurity and Data Science

- Coursework: Compiler Design and Development, AI and Machine Learning in High Performance Computing, Network Architecture and Security, Theory of Computing, Software Design Practices

TECHNICAL SKILLS

Languages: C, C++, Python, Go, Rust, Bash, TypeScript, Java, SQL

Infrastructure & Platform: Docker, Linux/Unix, Git, CI/CD

Data & Backends: MySQL, PostgreSQL, Redis

Security: AES / RSA / ECDH, JWT / auth

INTERESTS

Aviation (flying small aircraft), rock climbing